

# Topic 2



## Lecture 2 · Exploratory Data Analysis (EDA) Part 2

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National University of Singapore (NUS)

## TODAY'S OBJECTIVES

1

## Numerical Summaries of Center

- Mean
- Median

2

## Numerical Summaries of Variability

- Range
- Variance and Standard Deviation
- Interquartile Range (IQR)

# 01

## Numerical Summaries of Center

Mean and Median

# Notation for Samples

*Sample of  $n$  observations:*

$$X_1, X_2, X_3, \dots, X_n$$

Sample



Heights of 50 people

$X_1$ : Height of 1<sup>st</sup> person

$X_2$ : Height of 2<sup>nd</sup> person

...

$X_{50}$ : Height of 50<sup>th</sup> person

# Sample Mean



## DEFINITION

Sum of all observations in the sample divided by the sample size

## NOTATION

$$\bar{X}$$

## Formula

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \text{ n = sample size}$$

# Linear Transformation of Mean

For the purpose of this course, we will *only focus on one* linear transformation:

$$Y = bX + a$$

$b, a$  are known constants

Sample of  $n$  observations:

$$\begin{aligned} Y_1 &= bX_1 + a \\ Y_2 &= bX_2 + a \\ &\vdots \\ Y_n &= bX_n + a \end{aligned}$$

New sample

$$\bar{Y} = \frac{Y_1 + Y_2 + \dots + Y_n}{n}$$

$$\bar{Y} = b\bar{X} + a$$

# Linear Transformation and the Mean

## Example 1:

Heights of 10 people

$$\bar{X} = 1.65m$$

convert to cm



$$Y_i = 100X_i$$

$$\begin{aligned}\bar{Y} &= 100\bar{X} \\ &= 165 \text{ cm}\end{aligned}$$

## Example 2:

Celsius

°C

Average temperature in  
Singapore in July

$$\bar{X} = 29 \text{ °C}$$

Fahrenheit

°F

$$Y_i = \frac{9}{5}X_i + 32$$

$$\begin{aligned}\bar{Y} &= \frac{9}{5}\bar{X} + 32 \\ &= 84.2 \text{ °F}\end{aligned}$$

# Median



## DEFINITION

The middle value of the ordered observations. The median divides the sample points into 2 groups of equal size

## NOTATION

$$X_{(0.5)}$$

## EXAMPLE

If n is odd:  $X_{(0.5)} = X_{\frac{n+1}{2}}$

If n is even:  $X_{(0.5)} = \text{average of } X_{\frac{n}{2}} + X_{\frac{n}{2}+1}$

**9 observations:** 152, 154, 155, 159, 160, 160, 167, 172, 174

$$X_{(0.5)} = 159$$

**10 observations:** 150, 152, 154, 155, 159, 160, 160, 167, 172, 174

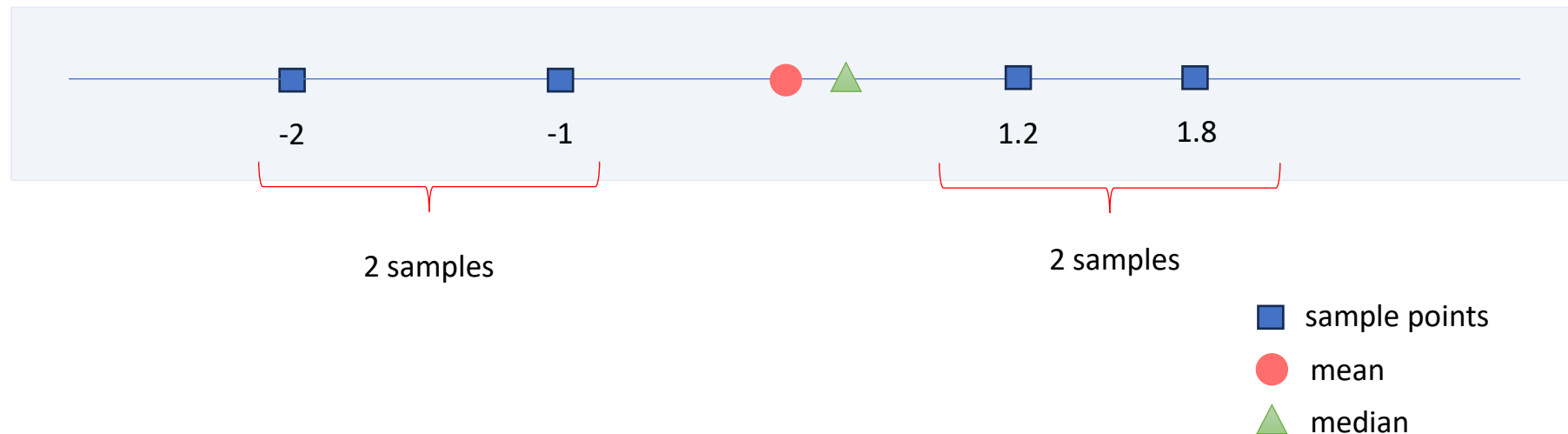
$$X_{(0.5)} = \frac{159+160}{2} = 159.5$$

# Comparison of Mean and Median

Sample of 4 observations: -2, -1, 1.2, 1.8

$$\bar{X} = \frac{-2 + (-1) + 1.2 + 1.8}{4} = 0$$

$$X_{(0.5)} = \frac{-1 + 1.2}{2} = 0.1$$

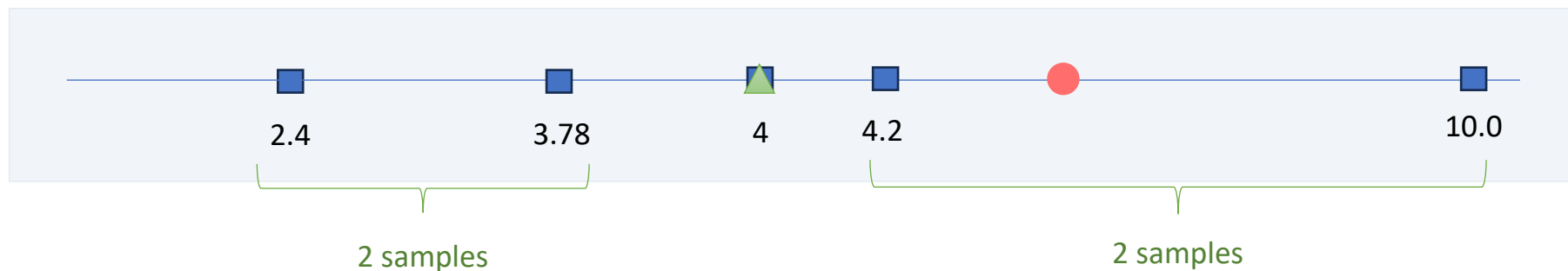


# Comparison of Mean and Median

Sample of 5 observations: 2.3, 3.78, 4, 4.2, 10.0

$$\bar{X} = \frac{4 + 2.3 + 3.78 + 10.0 + 4.2}{5} = 4.856$$

$$X_{(0.5)} = 4$$



■ sample points

▲ median

● mean

# Effect of Extreme Observations

Sample of 5 observations: 2.3, 3.78, 4, 4.2, **100**

$$\bar{X} = \frac{2.3+3.78+4+4.2+100}{5} = 22.856$$

Mean increases substantially

$$X_{(0.5)} = 4$$

Median remains unchanged



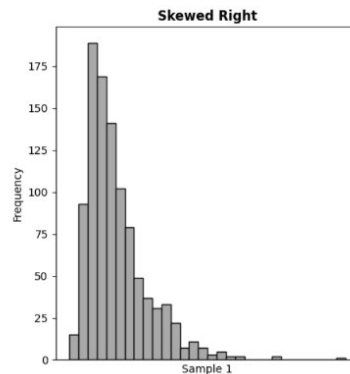
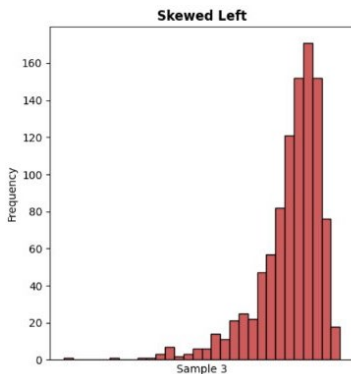
## KEY INSIGHT

**Mean:** depends on the ordering of observations

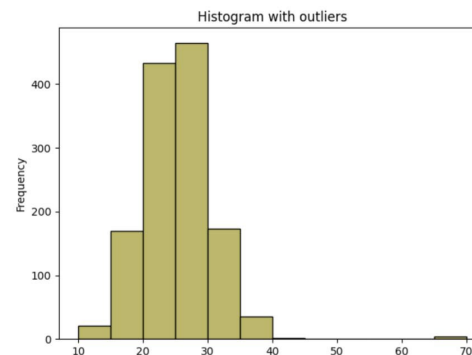
**Median:** depends on the actual values of all observations

# Comparison of Mean and Median

## Highly-skewed



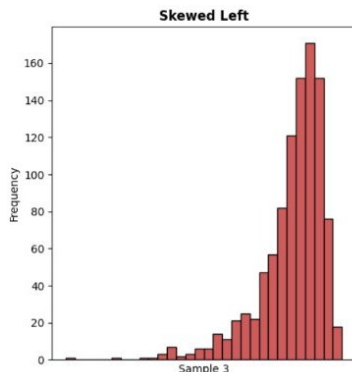
## Distribution with outliers



The median is a more representative summary of the center of distribution

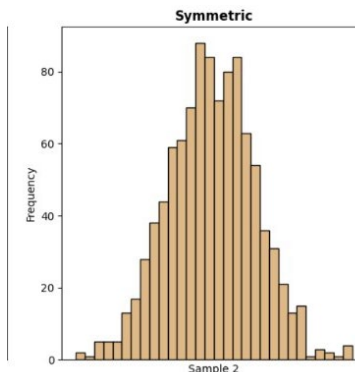
# Comparison of Mean and Median

Left-skewed



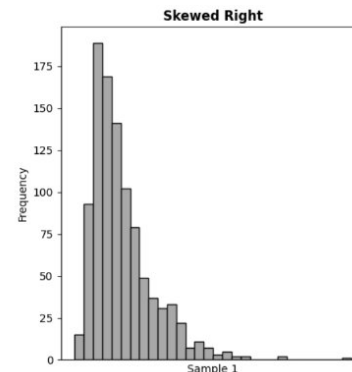
- Mean is **typically smaller** than median

Symmetric



- Mean and median are **approximately equal**

Right-skewed



- Mean is **typically larger** than median

# Comparison of Mean and Median

## Mean

**Balance point** of a distribution

**Influenced by** extreme observations

## Median

Middle observation that divides the sample points into **2 groups of equal size**

**Resistant to** extreme observations

Answers: what is a **typical observation** in a dataset?

Both reduce the dataset to a **simple summary**, potentially **hiding important details** of the distribution

# 02

## Variability

Range, Variance and Standard Deviation, IQR

# Variability



## DEFINITION

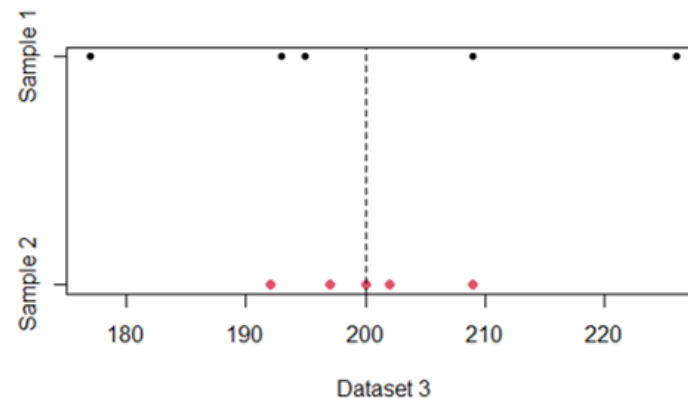
Describes how much the observations differ from one another and how far they tend to be from the centre of the distribution

## Note:

Variability is also known as **spread** or **dispersion**

## Importance of Variability:

2 datasets can have exactly the same mean but look very different.



# Range



## DEFINITION

The difference between the largest and the smallest observations in that dataset

Range is highly sensitive to extreme observations

Heights of 5 adults:

Sample 1  $X_1 = 1.55, X_2 = 1.61, X_3 = 1.65, X_4 = 1.70, X_5 = 1.90$

Range =  $1.90 - 1.55 = 0.35$

Sample 2  $X_1 = 1.55, X_2 = 1.61, X_3 = 1.65, X_4 = 1.70, X_5 = 2.25$

Range =  $2.25 - 1.55 = 0.70$

**Range  
doubled**

# Variance and Standard Deviation

Variance,  $s^2$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Units are **squared**

Standard Deviation,  $s$

$$s = \sqrt{s^2}$$

**Same units** as the original observations

Measure of **how far** observations typically lie **from the mean**

Sample 1  $X_1 = 177, X_2 = 193, X_3 = 195, X_4 = 209, X_5 = 226$

$$s^2 = 340$$

$$s = 18.4$$

**widely  
spread out**

Sample 2  $X_1 = 192, X_2 = 197, X_3 = 200, X_4 = 202, X_5 = 209$

$$s^2 = 39.5$$

$$s = 6.3$$

**concentrated  
around mean**

# Linear Transformation of Variance

Sample of  $n$  observations:

$$X_1, X_2, X_3, \dots, X_n$$

Standard deviation:  $s_x$

Variance:  $s_x^2$

Apply Transformation

$$Y_i = bX_i + a \quad \text{①}$$

$b, a$  are constants

$$s_y^2 = b^2 s_x^2 \quad \text{②}$$

$$s_y = |b| s_x \quad \text{③}$$

$$Y_1, Y_2, Y_3, \dots, Y_n$$

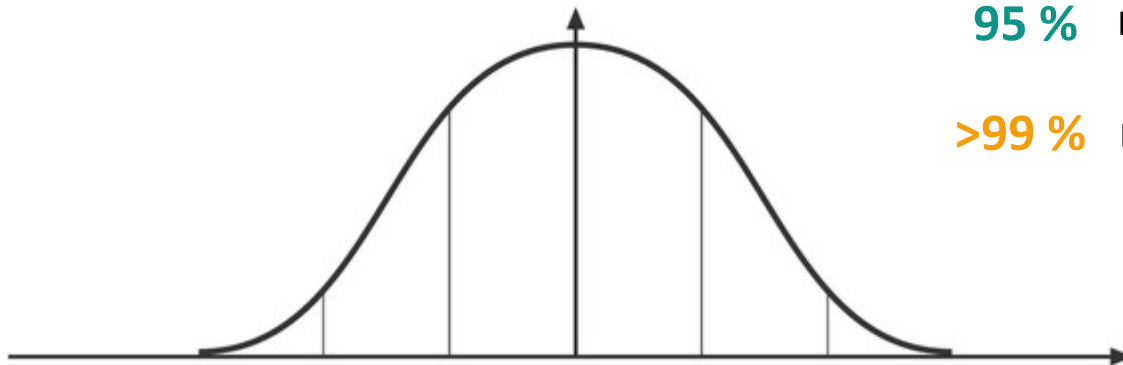
Standard deviation:  $s_y$

Variance:  $s_y^2$

- ① Adding a constant does not change variability.
- ② Multiplying a constant changes variability by  $b^2$ .
- ③ Standard deviation cannot be negative

# Empirical Rule

Applies to distributions approximately normal:

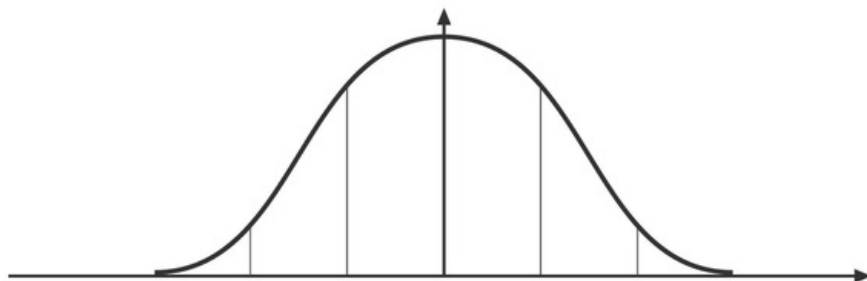


**68 %** Lie within 1 standard deviation of the mean

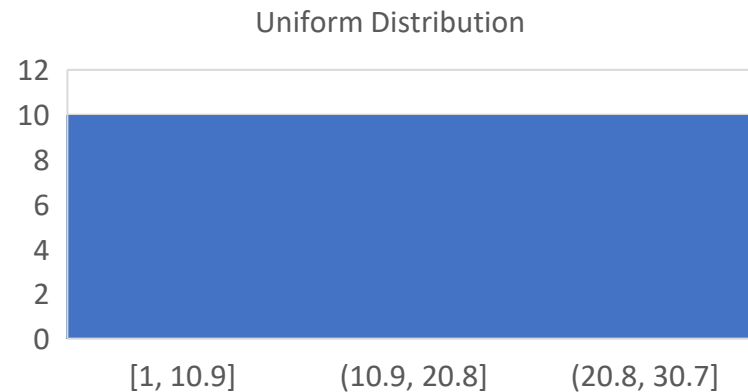
**95 %** Lie within 2 standard deviations of the mean

**>99 %** Lie within 3 standard deviations of the mean

# Normal vs Symmetric



**A normal distribution is always symmetric**



**A symmetric distribution is **not always** normal**

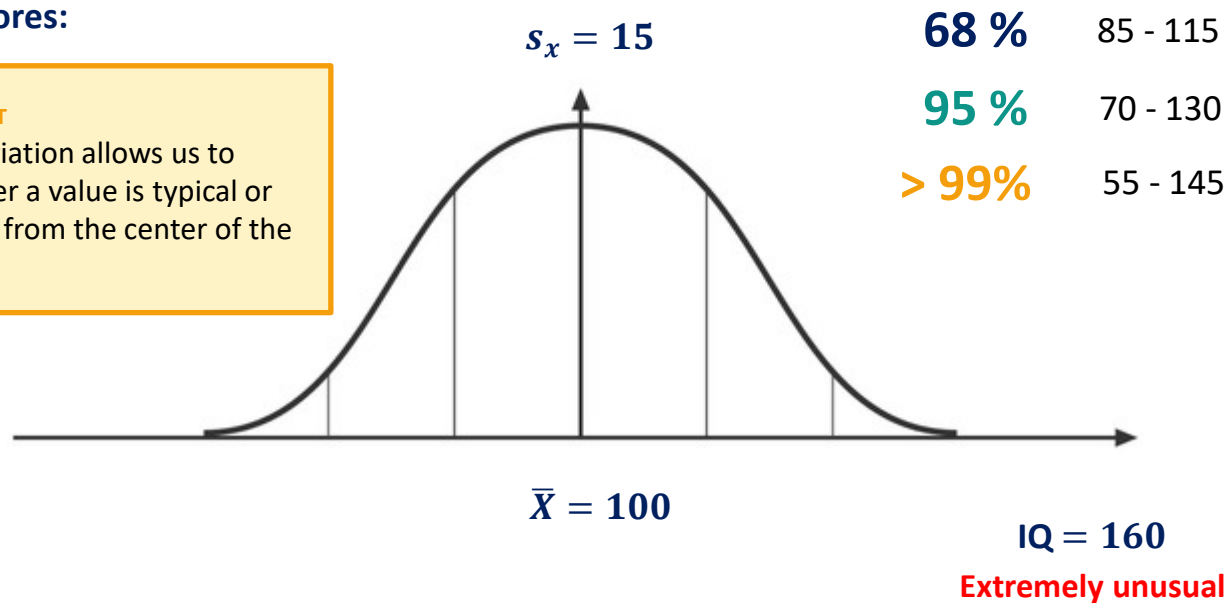
# Empirical Rule

## Distribution of IQ scores:



### KEY INSIGHT

Standard deviation allows us to judge whether a value is typical or unusually far from the center of the distribution.



# Quantile

Let  $p$  be a number between 0 and 1



## DEFINITION

The 100p-th quantile, is a value such that approximately 100p% of the observations are less than or equal to that value. Quantiles are also known as percentiles.

## Notation:

$$q_p$$

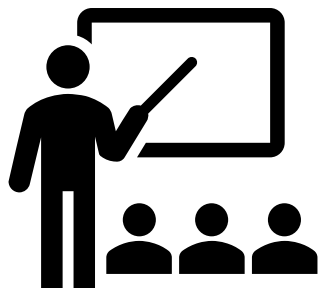
## Example:

Sample of 100 observations: 1, 2, 3, ..., 100

90% of the values are less than or equal to 90

$$q_{0.90} = 90 = 90^{\text{th}} \text{ percentile}$$

# Quantiles Example 1



Class of 50 students

Exam Score: 84

**Only one** student scored higher than you

**What percentile is your score?**

**49 students scored less than or equal to 84**

**percentile =  $\frac{49}{50} = 98^{\text{th}}$  percentile**

# Quantiles

Quantiles are not always uniquely defined

Sample of 200 observations:

1, 2, 3, ..., 200

$q_{0.25}$  = 25% of observations are

50



50.5



50-51



# Quartile

 $Q_1$ 

1<sup>st</sup> Quartile

- 25% of the observations are at or below this value.
- Also known as lower quartile.

 $Q_2$ 

2<sup>nd</sup> Quartile

- 50% of the observations are at or below this value. It is  $X_{0.5}$

 $Q_3$ 

3<sup>rd</sup> Quartile

- 75% of the observations are at or below this value.
- Also known as upper quartile

# Interquartile Range (IQR)



## DEFINITION

Distance between the first and third quartile.

## Formula

$$Q_3 - Q_1$$

## Note

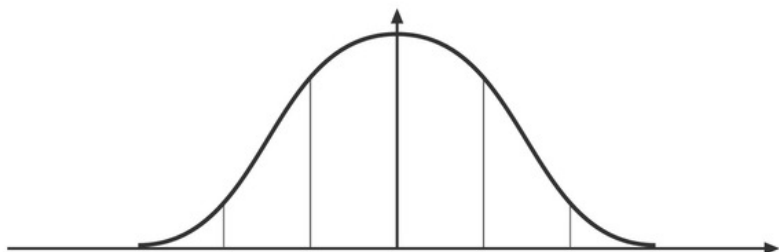
- 50% of the sample points fall within the IQR
- IQR measures the spread of the middle 50% of the observations.

Sample of 200 observations: 1, 2, 3, ..., 200

$$Q_1 = 50 \quad Q_3 = 150 \quad IQR = Q_3 - Q_1 = 100$$

# Choice of Numerical Summary

Bell-shaped, Symmetric

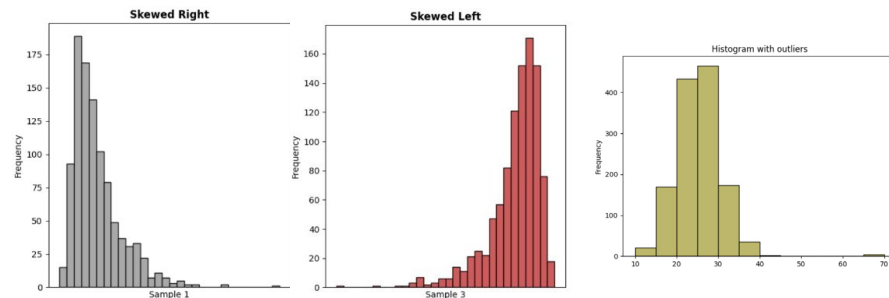


Mean

Standard Deviation

Range

Skewed / Contains outliers



Median

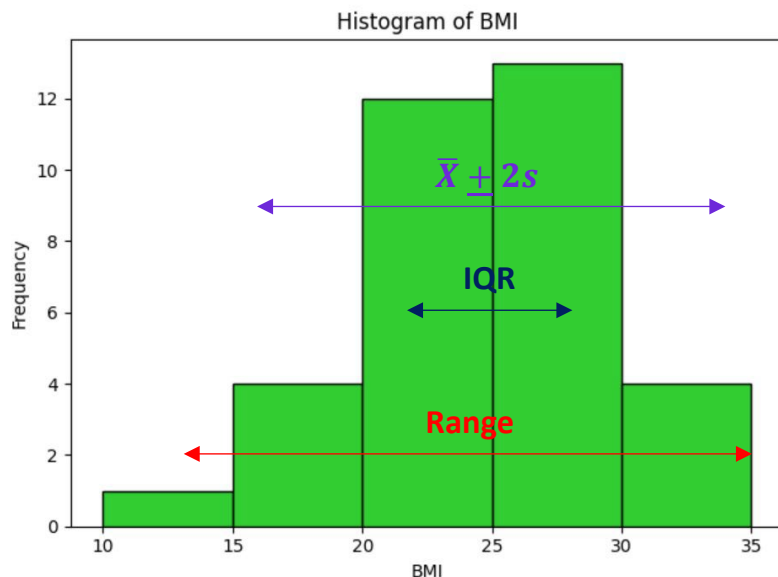
Interquartile Range (IQR)

# Descriptive Measures for BMI Variable

| Min. | 1 <sup>st</sup> Q | Median | Mean      | 3rd Q | Max  | Var       | sd       |
|------|-------------------|--------|-----------|-------|------|-----------|----------|
| 14.0 | 22.325            | 25.25  | 24.905882 | 28.45 | 35.0 | 22.744207 | 4.769089 |

$$\text{Range} = 35 - 14 = 21$$

$$\text{IQR} = 28.45 - 22.33 = 6.12$$



**Range: encompasses 100% of the data points**

**IQR: encompasses 50% of the data points**

**$\bar{X} \pm 2s$ : encompasses approximately 95% of the data points**

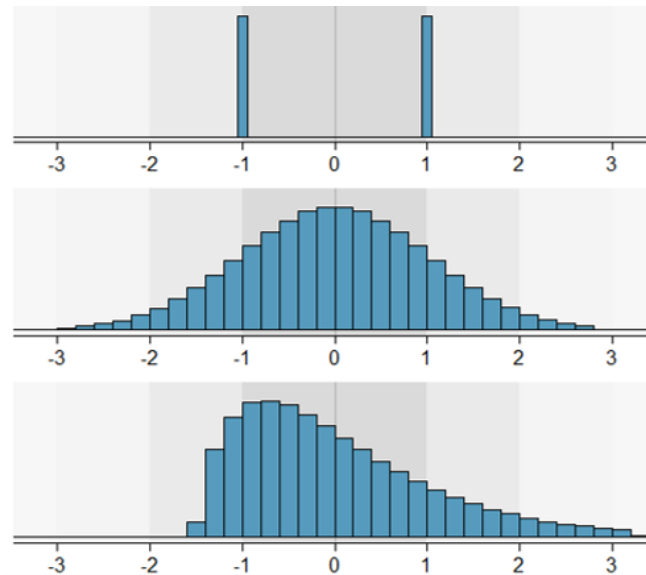
The **larger** these intervals are, the **more variability** there is in the data, and the **less peaked** the histogram will be.

# Numerical Summaries Are Not Enough

All 3 samples have:

$$\text{Mean} = 0$$

$$\text{Variance} = 1$$



No collection of numerical summaries can fully replace a graphical display of the data